

CIS — Nova Integration Matrix v1.0

Cross-system integration map: Nova Bank Online ↔ Nova 1 Chain 22016

Field	Value
Document ID	CIS-NOVA-INTEGRATION-MATRIX-v1
Version	1.0
Components	Nova Bank Online + Nova 1 Chain (22016)

1. Component summary

Component	CIS document	String ID	Key identifier
Nova Bank Online	CIS-Nova-Bank-Online-v1.md	nsb / nova	SWIFT NSBKLAL2X
Nova 1 Chain	CIS-Nova-1-Chain-22016-v1.md	nova-1	Chain ID 22016 (0x5600)
OneX Bridge	(platform)	onex-production-platform	Port 9338

2. Integration matrix

#	Source	Target	Flow	API / mechanism	Settlement kind
1	Nova Bank account	Nova Bank account	Internal transfer	POST /bridge/bank/transfer	internal
2	Nova Bank M1	External IBAN	SEPA / SWIFT / wire	POST /bridge/bank/transfer + rail	real_fiat
3	Nova Bank ledger	Nova 1 wallet	Fiat → NOVA payout	POST /bridge/ledger/settle	real_crypto
4	Nova 1 wallet	Nova Bank account	Crypto → fiat (via HYBX)	POST /bridge/bank/hybx/exchange	hybx-nova-1
5	Fineract core	Nova Bank Online	Account sync	POST /bridge/bank/fineract/sync	—
6	HYBX exchange	Nova Bank + Nova 1	Multi-ledger routing	POST /bridge/bank/hybx/exchange	middleware
7	Bridge7 ledgers	Real ledger	Multi-book merge	POST /bridge/bridge7/sync	—
8	Virtual card spend	Nova Bank debit	Card authorization	/bridge/cards/*	internal debit
9	Token platform	Nova 1 ERC-20	Deploy / wrap	POST /bridge/platform/deploy	on-chain
10	Nova Bank M0/M1/NSB	Liquidity pool	Fund class routing	Fiat settlement middleware	vault / pool
11	Card payer (Visa/MC/Amex)	Nominated bank account	Donation / payment / collection	POST /bridge/payments/* + /payments/ portal	internal or real_fiat

3. End-to-end flows

3.1 Nova Bank → Nova 1 Chain (fiat to crypto)

[Architecture diagram — see markdown source]

```
sequenceDiagram
    participant User
    participant Wallet as Wallet UI
    participant Bridge as OneX Bridge
    participant Ledger as Real Ledger
    participant EVM as Nova 1 RPC 22016

    User->>Wallet: Settle to Nova 1 address
    Wallet->>Bridge: POST /bridge/ledger/settle
    Note over Bridge: fromAccount: nova-usd-checking<br/>externalTo: nova-1:0x...
    Bridge->>Ledger: Debit M1/NSB account
    Bridge->>EVM: eth_sendRawTransaction
    EVM-->>Bridge: tx hash
    Bridge-->>Wallet: settlement completed
```

Request:

```
POST /bridge/ledger/settle
{
  "fromAccount": "nova-usd-sovereign",
  "amount": "1000",
  "payoutAsset": "NOVA",
  "kind": "real_crypto",
  "externalTo": "nova-1:0xRecipientAddress"
}
```

3.2 Nova 1 → Nova Bank (crypto to fiat via HYBX)

```
POST /bridge/bank/hybx/exchange
{
  "route": "nova-1-hybx",
  "from": "nova-1:0xSenderAddress",
  "to": "nova-usd-checking",
  "amount": "50",
  "asset": "NOVA"
}
```

3.3 Cross-chain wrap (Nova 1 ↔ other EVM chains)

```
POST /bridge/platform/wrap
{
  "fromChain": "ethereum",
  "fromToken": "USDT",
  "toChain": "nova-1",
  "amount": "1000"
}
```

4. Shared configuration

Deploy both components on the same bridge instance:

```
# Nova Bank Online
ONEX_LEDGER_MODE=production
ONEX_ONLINE_BANK=1
ONEX_BANK_LEDGER_FILE=configs/bank-ledger.nova.example.json
ONEX_HYBX_ENABLED=1
ONEX_FINERACT_ENABLED=1

# Nova 1 Chain 22016
ONEX_DEFAULT_BRIDGE_CHAIN=nova-1
NOVA1_RPC_URL=https://rpc.nova1.chain
NOVA1_CHAIN_ID=22016
ONEX_EVM HOLDER=0xTreasury0nNova1
# ONEX_EVM_SENDER_KEY=<funded-signer>

ONEX_API_KEY=<secret>
```

5. Unified verification

```
# Platform
curl -s https://HOST/bridge/production/status | jq '{onlineBank, defaultBridge: .ledger.defaultBridgeChain}'

# Nova Bank
curl -s https://HOST/bridge/bank/status | jq '{name, online, accounts: .accountCount}'

# Nova 1 Chain
curl -s https://HOST/bridge/ledger/status | jq '{defaultBridgeChain, nova1Rpc: .nova1Configured}'

# HYBX routes
curl -s https://HOST/bridge/bank/hybx/exchange/routes | jq .
```

6. Fund class routing (Nova Bank → settlement)

Fund class	Typical source account	Nova 1 settlement
m0	nova-usd-cash (reserves)	Routed via M1 pool before crypto payout
m1	nova-usd-checking	Direct real_crypto settlement
nsb	nova-usd-sovereign	Direct real_crypto settlement

7. Document index

Document	Path
Nova Bank Online CIS	docs/cis/CIS-Nova-Bank-Online-v1.md
Nova 1 Chain 22016 CIS	docs/cis/CIS-Nova-1-Chain-22016-v1.md
Nova deploy (chain)	deploy/DEPLOY-nova-1-22016.md
Nova deploy (trustee)	deploy/DEPLOY-novatruster.digital.md
Bank ledger seed	configs/bank-ledger.nova.example.json
Chain registry	configs/chains.json
HYBX middleware	configs/hybx-middleware.example.json

